

June 13, 2026

Editor-in-Chief
Electronics
MDPI

Dear Editor,

I am pleased to submit our manuscript entitled “Deep Learning for Student Engagement Detection: A Lightweight CNN Approach for Educational Environments” for consideration for publication in *Electronics*.

This paper proposes a lightweight deep learning framework integrating MobileNetV3 with a novel Channel-Spatial Attention Module (CSAM) for real-time student engagement recognition on edge devices. The key contributions are:

1. A lightweight CSAM module optimized for mobile architectures, achieving 1.3–5.0% accuracy improvement with only 0.01M additional parameters and negligible computational overhead.
2. A comprehensive evaluation across three datasets (FER2013, CK+, DAiSEE) with cross-dataset generalization and ablation studies.
3. Deployment-oriented evaluation including parameter count, FLOPs, inference latency, and energy consumption on GPU, demonstrating suitability for edge deployment.

The MobileNetV3+CSAM model achieves 70.5% accuracy with only 1.26M parameters, 0.068G FLOPs, and 53 fps inference speed, making it well-suited for resource-constrained embedded systems in educational settings. This work is directly relevant to the scope of *Electronics* as it addresses efficient deep learning inference on edge hardware.

Both authors have read and approved the manuscript. The work has not been published elsewhere and is not under consideration by another journal. We declare no conflicts of interest.

I look forward to your decision.

Sincerely,

Huanjie Yu
Hunan University of Technology and
Business
Email: semhaqx@gmail.com